

Exercises 4 and 5 - Theoretical questions

Exercise 4 - Theoretical question 1

Describe the basic approximations used in the Hartree-Fock Self Consistent Field Method: Born-Oppenheimer Approximation, Orbital Approximation, Mean Field Approximation. How do these approximations simplify the computational problem?

Born-Oppenheimer. Nuclei are about 1836 times heavier than an electron, so on the timescale of nuclear motion the electrons relax essentially instantly to their ground state at any fixed nuclear geometry. The total wavefunction is then separated as $\Psi = \Psi_{\text{el}}(\mathbf{r}; \mathbf{R}) \cdot \chi_{\text{nuc}}(\mathbf{R})$, with $\mathbf{R}$ entering the electronic problem only as a parameter. Solving $\hat{H}_{\text{el}}\Psi_{\text{el}} = E_{\text{el}}(\mathbf{R})\Psi_{\text{el}}$ once per fixed geometry, instead of together with the nuclei, turns one coupled problem into a 3N-electron electronic problem plus a potential energy surface $E_{\text{el}}(\mathbf{R}) + V_{\text{nn}}(\mathbf{R})$ that can be optimised, scanned or differentiated afterwards.

Orbital approximation. The N-electron wavefunction is approximated by a single Slater determinant of N one-electron spin orbitals, $\Psi = (1/\sqrt{N!}) \cdot \det[\chi_1(x_1) \dots \chi_N(x_1); \dots; \chi_1(x_N) \dots \chi_N(x_N)]$, which builds in the Pauli principle automatically (swapping two electrons flips the sign; two electrons in the same spin orbital give a zero determinant). One unknown function of 3N coordinates becomes N unknown one-electron functions of 3 coordinates each, and each of those can be expanded in a finite atomic-orbital basis - so finding the orbitals is now a linear-algebra (matrix diagonalisation) problem rather than a differential equation.

Mean-field approximation. Each electron is treated as moving in the time-averaged charge distribution of the others rather than reacting to their instantaneous positions. This turns the electron-electron repulsion into an effective one-electron Fock operator, $f(1) = h(1) + \sum_j [J_j(1) - K_j(1)]$, $f\phi_i = \epsilon_i\phi_i$, so the N-electron problem is replaced by N one-electron eigenvalue problems, solved iteratively until self-consistent (SCF) because f depends on the orbitals it produces. This is far cheaper than the exact many-body problem, at the price of leaving out electron correlation - the instantaneous part of the electron-electron interaction.

Exercise 5 - Theoretical question 2

Describe the covalent chemical bonding on the example of the H₂ molecule. Write down the Slater Determinant for the ground state and give an estimate of the total energy using the method explained in Lecture 3 (with h , J , K). Write down the energies of all excited Slater Determinants that can be formed using the minimal basis set on each of the H atoms. Explain the Coulomb and Exchange integrals.

Bonding picture. Each H atom contributes one 1s orbital, a on atom A and b on atom B, with overlap $S = \langle a | b \rangle$. Symmetry gives a bonding and an antibonding combination:

$$\sigma_g = \frac{a + b}{\sqrt{2(1 + S)}} \quad \sigma_u = \frac{a - b}{\sqrt{2(1 - S)}}$$

σ_g piles electron density up between the nuclei (lowering the energy - the covalent bond); σ_u has a node there and lies higher in energy.

Ground-state determinant and energy. Both electrons occupy σ_g with opposite spin:

$$\psi_0 = \frac{|\sigma_g(\alpha) \sigma_g(\beta)|}{\sqrt{2}}$$

With $E = \sum_i h_{ii} + \sum_{i < j} (J_{ij} - K_{ij}) + V_{nn}$ and only one opposite-spin pair ($K = 0$):

$$E_0 = 2h_{gg} + J_{gg} + V_{nn}$$

$h_{gg} = (h_{aa} + h_{ab})/(1+S)$ and J_{gg} is the Coulomb self-repulsion of the σ_g charge cloud. The resonance integral $h_{ab} = \langle a | \hat{h} | b \rangle$ is negative and sizeable - it is what pulls E_0 below the energy of two separated atoms ($2h_{aa}$) and gives the covalent bond.

Excited determinants. With 2 spatial orbitals (σ_g, σ_u) there are 4 spin orbitals, so besides the ground configuration there are 5 more determinants:

Determinant	M_S	Energy
$\sigma_g(\alpha) \sigma_g(\beta)$ (ground state)	0	$2h_{gg} + J_{gg} + V_{nn}$
$\sigma_g(\alpha) \sigma_u(\alpha)$	+1	$h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$
$\sigma_g(\beta) \sigma_u(\beta)$	-1	$h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$
$\sigma_g(\alpha) \sigma_u(\beta)$	0	$h_{gg} + h_{uu} + J_{gu} + V_{nn}$
$\sigma_g(\beta) \sigma_u(\alpha)$	0	$h_{gg} + h_{uu} + J_{gu} + V_{nn}$
$\sigma_u(\alpha) \sigma_u(\beta)$ (doubly excited)	0	$2h_{uu} + J_{uu} + V_{nn}$

The two $M_S = 0$ singly-excited determinants are not eigenstates by themselves; their symmetric and antisymmetric combinations give the spin-adapted triplet (energy $h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$) and singlet ($h_{gg} + h_{uu} + J_{gu} + K_{gu} + V_{nn}$) excited states.

Coulomb and exchange integrals. Coulomb integral:

$$J_{ij} = (ii | jj) = \iint \phi_i(1)^2 (1/r_{12}) \phi_j(2)^2 d\tau$$

This is the classical electrostatic repulsion between the charge clouds $|\phi_i|^2$ and $|\phi_j|^2$: always positive, and independent of the relative spin of the two electrons.

Exchange integral:

$$K_{ij} = (ij|ji) = \iint \phi_i(1)\phi_j(1) (1/r_{12}) \phi_j(2)\phi_i(2) d\tau$$

This term has no classical counterpart - it comes purely from the antisymmetry of the Slater determinant. It is non-zero only between same-spin electrons and enters the energy with a minus sign, which is why same-spin (triplet) configurations sit lower than opposite-spin ones: the quantum origin of Hund's rule.